PCB Physics Reference

Trace width and length · Ground planes · Routing · Component placement · Heat

This reference covers what happens when ideal circuit diagrams meet physical copper: why trace width is chosen for current and clearance for voltage, why ground carries just as much current as the supply rail, how routing and component placement affect a board, and how heat moves through it. Worked examples use the 1-LED project (5 V, 220 Ω, ~13.6 mA) so the numbers stay concrete.

> **Key idea:** The single most important rule of this whole document: trace WIDTH is chosen for CURRENT; trace CLEARANCE (spacing) is chosen for VOLTAGE.

1 — Trace width: a trace is a small resistor

1.1 Copper has resistance

A PCB trace is not an ideal wire. It is a thin strip of copper, and copper has resistivity. Every trace is therefore a small resistor obeying:

$$R = \rho \times L \div A$$

where ρ (rho) is copper's resistivity (1.68×10^{-8} Ω·m), L is the trace length, and A is the cross-section area = width × copper thickness. Standard PCB copper ("1 oz") is 35 µm (0.035 mm) thick — that thickness is fixed when you order the board, so **width is the knob you control** in the layout.

1.2 Why current sets the width

When current I flows through the trace's resistance R, the trace dissipates $P = I^2 \times R$ as heat. Notice the square: doubling the current quadruples the heating. A trace that is too narrow for its current warms up, its resistance rises further (copper resistance grows with temperature), and in extreme overload it acts as an unintentional fuse and burns open. Width is therefore chosen so the trace stays acceptably cool at its working current. Rough guide for 1 oz copper, external layer, ~10 °C of allowed self-heating:

Trace width	Comfortable continuous current
0.15 mm (minimum at many fabs)	~0.5 A
0.25 mm (typical signal trace)	~0.7 A
0.5 mm	~1.4 A
1.0 mm	~2.3 A
2.0 mm	~4 A
Copper plane / pour	many amps

These values come from the IPC-2221 industry charts (approximate — use a trace-width calculator for critical designs). For the 1-LED board at 13.6 mA, even the thinnest manufacturable trace is loaded to barely 2 % of its capacity: nothing on that board is width-critical. 0.25–0.5 mm is a comfortable choice.

1.3 Why voltage sets the clearance, not the width

Voltage does not care how wide a trace is — 5 V travels down a hair-thin trace as happily as a wide one. What a high voltage can do is jump or leak *across the gap to a neighboring trace*. The required gap grows with voltage: at 5 V any manufacturable spacing (≥ 0.15 mm) is already far more than needed, while mains voltage (110–230 V AC) demands several millimeters of separation ("clearance" through air, "creepage" along the board surface) and is governed by safety standards. Until you design for high voltage, the fab's default rules cover you.

Key idea: Current decides how WIDE a trace must be (heating). Voltage decides how FAR APART traces must be (arcing/leakage). Two different knobs for two different quantities.

2 — Trace length: resistance you walk along

2.1 Length adds resistance linearly

From $R = \rho \times L \div A$: double the length, double the resistance. The trace then steals a small voltage drop ($V = I \times R$) from your budget — exactly like a tiny extra series resistor in the Kirchhoff sum. Worked example for a realistic trace on the LED board — 0.25 mm wide, 1 oz (0.035 mm) copper, 50 mm long:

$$R = (1.68 \times 10^{-8} \times 0.05) \div (0.00025 \times 0.000035) \approx 0.1\ \Omega$$

$$V_{lost} = I \times R = 0.0136\ \text{A} \times 0.1\ \Omega \approx 1.4\ \text{mV}$$

1.4 millivolts out of a 5000 mV budget — utterly negligible, which is why trace length can be ignored for small DC circuits and you may route wherever convenient.

2.2 When length starts to matter

High current: at 5 A, that same 0.1 Ω trace would drop 0.5 V and dissipate 2.5 W — a hot, failing trace. High-current paths must be short and wide (or planes).

High speed / high frequency: a long trace is an antenna and a transmission line. Fast digital edges reflect off the ends of long traces, pick up and radiate noise, and signal arrival time itself starts to matter (signals travel ~15 cm per nanosecond in PCB material). These effects appear with microcontrollers, USB, and faster — a later chapter.

Precision analog: millivolts matter when measuring sensors, so sense lines are kept short or use dedicated techniques (Kelvin connections).

3 — The B.Cu ground plane: 0 V is not zero current

3.1 The trap: "it's only ground, so it's fine"

The return side of the circuit carries **exactly the same current** as the supply side. Every milliamp that leaves +5 V must come back through GND — current flows in a loop, always. "0 V" describes the potential we measure from, not the amount of traffic. On the 1-LED board, the GND copper carries the full 13.6 mA, no less than the +5 V trace does.

3.2 Ground bounce — when GND is not 0 V

Because the return copper has resistance too, the return current drops a small voltage along it ($V = I \times R$, as always). The consequence: the point the LED's cathode calls "GND" actually sits a few millivolts *above* the supply's true 0 V. Engineers call this ground bounce or ground offset. At 13.6 mA it is harmless; in circuits mixing heavy currents with sensitive measurements, a bouncing ground corrupts every reading — which is why return paths get serious attention in professional layouts.

3.3 Why pouring the whole back layer as GND is good practice

A copper pour (plane) covering the entire B.Cu layer is effectively a trace of enormous width — so its resistance is close to zero and the return current barely drops any voltage at all. Three further free benefits: the plane **spreads heat** across the whole board (copper conducts heat ~1000× better than the fiberglass); it acts as an **electrical shield**, soaking up noise instead of letting traces radiate or receive it; and it makes routing easier, since any component can reach GND with a single via instead of a long trace. So yes — flood B.Cu with ground. Just know it is a hard-working conductor, not an idle 0 V backdrop.

> **Key idea:** GND carries the full loop current. A ground plane is excellent practice because it is a near-zero-resistance return path, a heat spreader, and a noise shield — not because "0 V doesn't matter."

3.4 Return current follows the path of least impedance

A subtlety to file away for later: in a plane, the return current does not wander randomly. At DC it spreads to take the lowest-resistance path; at high frequencies it hugs the area directly underneath the signal trace above it, minimizing the loop. This is why cutting slots in a ground plane under signal traces is a classic layout mistake in fast designs — it forces the return current on a detour, enlarging the loop and creating noise.

4 — How routes behave

4.1 Short and direct wins

Every centimeter of trace adds resistance, inductance, and antenna area. The default instinct should be: shortest reasonable path. You do not need to optimize obsessively on simple boards — but never make a trace long for decoration.

4.2 Corners: 45° by convention

Use two 45° bends instead of one sharp 90° corner, and never acute (< 90°) angles. The electrical benefit at low speed is negligible — the real reasons are manufacturability (acute inside corners can trap etching chemicals, the classic "acid trap") and convention: boards routed at 45° read as professional work. KiCad's router does this for you by default.

4.3 Loop area: the hidden antenna

The signal trace and its return path together form a loop, and the **area** of that loop determines how well it works as an antenna — both for transmitting noise and receiving it. Keep outgoing and return paths close together (a ground plane directly under the signal layer achieves this automatically). This single principle explains a large fraction of all professional layout rules.

4.4 Vias: stairs between floors

A via is a small plated hole connecting layers — how a front-side trace reaches the back-side ground plane. Each via adds a tiny resistance (~0.5 mΩ) and inductance; irrelevant on simple boards, counted carefully on fast or high-current ones. Multiple vias in parallel share current and conduct heat between layers.

4.5 Trace-to-trace spacing

Beyond the voltage clearance of section 1.3, two traces running side by side for a long distance couple capacitively and inductively — a fast signal on one induces a ghost copy on the other ("crosstalk"). Rule of thumb when it matters: keep spacing at least 3× the trace width for sensitive parallel runs. Irrelevant for the LED board; vital around microcontrollers and analog sensors.

5 — Components near each other

5.1 Close is mostly good

Placing connected components close together shortens traces — less resistance, less loop area, less noise. Professional layout begins by placing components in the order the signal flows, then routing falls out naturally. The classic later rule: decoupling capacitors must sit within millimeters of the chip pin they serve, because their entire job depends on a tiny loop.

5.2 The three cautions

(1) Soldering and rework room. Your iron needs physical access; parts touching each other are miserable to assemble and impossible to replace. Leave at least a part's-width of breathing space around anything you might rework.
(2) Thermal coupling. A hot component heats its neighbors through the board. The classic victim is the electrolytic capacitor, whose lifespan roughly *halves for every 10 °C* of extra temperature — never place one against a regulator or power resistor.
(3) Electrical interference. Noisy parts (switching regulators, motor drivers, crystals) are kept away from sensitive ones (analog sensors, radio sections). On the LED board nothing is noisy and nothing is sensitive — but the habit of asking "who is loud, who is listening?" starts now.

5.3 Mechanical reality

Connectors go at board edges where cables can reach them; screw terminals need screwdriver access from above; mounting holes need clearance for screw heads; tall components must not block them. Sketching the enclosure first is never wasted time.

6 — How heat behaves on a board

6.1 Every component is a small heater

Any component dropping voltage while carrying current converts $P = V \times I$ into heat, continuously. The LED board produces ~41 mW in the resistor and ~27 mW in the LED — about 68 mW total, less than a thousandth of a small light bulb. It will not get measurably warm. A linear regulator dropping 7 V at 0.5 A, by contrast, makes 3.5 W — enough to burn a finger and destroy itself without help.

6.2 Copper is the highway, fiberglass is the wall

Heat leaves a component three ways: conduction into the board, convection into the air, radiation (minor at these temperatures). On the board itself, copper conducts heat about 1000× better than the FR4 fiberglass — so **copper area is your heatsink**. Wide traces, pours, and planes pull heat away from hot parts and spread it over the board surface, where the air can take it.

6.3 The toolkit for hot components

Technique	How it works
Wide traces / copper pour at the hot pad	More copper cross-section = heat highway
Thermal vias under the component	Rows of vias pipe heat to the plane on the other side
Ground plane	Whole-board heat spreader, for free
Spacing hot parts apart	Two heaters side by side share one patch of board
Derating	Use a part rated 2× the real load; it runs cool and lives long
Heatsink / airflow	When the board alone is not enough

6.4 Thermal relief — the soldering compromise

One practical quirk you will meet in KiCad: a pad connected directly to a large copper pour is hard to solder, because the pour sucks the iron's heat away faster than the joint can form. KiCad therefore connects pads to pours through *thermal relief spokes* — four thin bridges instead of solid copper. Slightly worse electrically and thermally, vastly better for your soldering iron. The default is right for hand-assembled boards.

6.5 Summary

Width for current, clearance for voltage. Length is a resistor you walk along — negligible at milliamps, decisive at amps and at high speed. The ground plane carries the full return current and is good practice precisely because it does that job with near-zero resistance, while spreading heat and blocking noise. Route short and direct with 45° bends and small loops. Place components close, but leave room for the iron, the heat, and the noise. And remember every $P = V \times I$ is a heater — copper is how it survives.